```sql
--*********************************************************************--
-- Title: Assignment06
-- Author: Mark Lorentzen
-- Desc: This file demonstrates how to use Views
-- Change Log: Answered questions 1 through 10 related to views.
-- 2026-03-04,Mark Lorentzen,Created File
--*********************************************************************--
Begin Try
    Use Master;
    If Exists(Select Name From SysDatabases Where Name = 'Assignment06DB_MLorentzen')
     Begin
      Alter Database [Assignment06DB_MLorentzen] set Single_user With Rollback Immediate;
      Drop Database Assignment06DB_MLorentzen;
     End
    Create Database Assignment06DB_MLorentzen;
End Try
Begin Catch
    Print Error_Number();
End Catch
Go
Use Assignment06DB_MLorentzen;

-- Create Tables (Module 01)--
Create Table Categories
([CategoryID] [int] IDENTITY(1,1) NOT NULL
,[CategoryName] [nvarchar](100) NOT NULL
);
Go

Create Table Products
([ProductID] [int] IDENTITY(1,1) NOT NULL
,[ProductName] [nvarchar](100) NOT NULL
,[CategoryID] [int] NULL
,[UnitPrice] [money] NOT NULL
);
Go

Create Table Employees -- New Table
([EmployeeID] [int] IDENTITY(1,1) NOT NULL
,[EmployeeFirstName] [nvarchar](100) NOT NULL
,[EmployeeLastName] [nvarchar](100) NOT NULL
,[ManagerID] [int] NULL
);
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Create Table Inventories
([InventoryID] [int] IDENTITY(1,1) NOT NULL
,[InventoryDate] [Date] NOT NULL
,[EmployeeID] [int] NOT NULL -- New Column
,[ProductID] [int] NOT NULL
,[Count] [int] NOT NULL
);
Go

-- Add Constraints (Module 02) --
Begin  -- Categories
    Alter Table Categories
     Add Constraint pkCategories
      Primary Key (CategoryId);

    Alter Table Categories
     Add Constraint ukCategories
      Unique (CategoryName);
End
Go

Begin -- Products
    Alter Table Products
     Add Constraint pkProducts
      Primary Key (ProductId);

    Alter Table Products
     Add Constraint ukProducts
      Unique (ProductName);

    Alter Table Products
     Add Constraint fkProductsToCategories
      Foreign Key (CategoryId) References Categories(CategoryId);

    Alter Table Products
     Add Constraint ckProductUnitPriceZeroOrHigher
      Check (UnitPrice >= 0);
End
Go

Begin -- Employees
    Alter Table Employees
     Add Constraint pkEmployees
      Primary Key (EmployeeId);

    Alter Table Employees
     Add Constraint fkEmployeesToEmployeesManager
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      Foreign Key (ManagerId) References Employees(EmployeeId);
End
Go

Begin -- Inventories
    Alter Table Inventories
     Add Constraint pkInventories
      Primary Key (InventoryId);

    Alter Table Inventories
     Add Constraint dfInventoryDate
      Default GetDate() For InventoryDate;

    Alter Table Inventories
     Add Constraint fkInventoriesToProducts
      Foreign Key (ProductId) References Products(ProductId);

    Alter Table Inventories
     Add Constraint ckInventoryCountZeroOrHigher
      Check ([Count] >= 0);

    Alter Table Inventories
     Add Constraint fkInventoriesToEmployees
      Foreign Key (EmployeeId) References Employees(EmployeeId);
End
Go

-- Adding Data (Module 04) --
Insert Into Categories
(CategoryName)
Select CategoryName
 From Northwind.dbo.Categories
 Order By CategoryID;
Go

Insert Into Products
(ProductName, CategoryID, UnitPrice)
Select ProductName,CategoryID, UnitPrice
 From Northwind.dbo.Products
  Order By ProductID;
Go

Insert Into Employees
(EmployeeFirstName, EmployeeLastName, ManagerID)
Select E.FirstName, E.LastName, IsNull(E.ReportsTo, E.EmployeeID)
 From Northwind.dbo.Employees AS E
  Order By E.EmployeeID;
Go

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Insert Into Inventories
(InventoryDate, EmployeeID, ProductID, [Count])
Select '20170101' AS InventoryDate, 5 AS EmployeeID, ProductID, UnitsInStock
From Northwind.dbo.Products
Union
Select '20170201' AS InventoryDate, 7 AS EmployeeID, ProductID, UnitsInStock + 10 -- Using this is to create a made up value
From Northwind.dbo.Products
Union
Select '20170301' AS InventoryDate, 9 AS EmployeeID, ProductID, UnitsInStock + 20 -- Using this is to create a made up value
From Northwind.dbo.Products
Order By 1, 2
Go

-- Show the Current data in the Categories, Products, and Inventories Tables
Select * From Categories;
Go
Select * From Products;
Go
Select * From Employees;
Go
Select * From Inventories;
Go

/********************************** Questions and Answers **********************************
NOTES
 1) You can use any name you like for you views, but be descriptive and consistent
 2) You can use your working code from assignment 5 for much of this assignment
 3) You must use the BASIC views for each table after they are created in Question 1
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-- Question 1 (5% pts): How can you create BASIC views to show data from each table in the database.
-- NOTES: 1) Do not use a *, list out each column!
--        2) Create one view per table!
--        3) Use SchemaBinding to protect the views from being orphaned!

Create View dbo.vCategories
       With SchemaBinding AS
       Select
            dbo.Categories.CategoryID,
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            dbo.Categories.CategoryName
        From
            dbo.Categories;
Go

Create View dbo.vProducts
       With SchemaBinding AS
       Select
            dbo.Products.ProductID,
            dbo.Products.ProductName,
            dbo.Products.CategoryID,
            dbo.Products.UnitPrice
       From
            dbo.Products;
Go

Create View dbo.vEmployees
       With SchemaBinding AS
       Select
            dbo.Employees.EmployeeID,
            dbo.Employees.EmployeeFirstName,
            dbo.Employees.EmployeeLastName,
            dbo.Employees.ManagerID
       From
            dbo.Employees;
Go

Create View dbo.vInventories
       With SchemaBinding AS
       Select
            dbo.Inventories.InventoryID,
            dbo.Inventories.InventoryDate,
            dbo.Inventories.EmployeeID,
            dbo.Inventories.ProductID,
            dbo.Inventories.[Count]
       From
            dbo.Inventories;
Go

-- Question 2 (5% pts): How can you set permissions, so that the public group CANNOT select data
-- from each table, but can select data from each view?

Deny Select On dbo.Categories to Public;
Grant Select On dbo.vCategories to Public;

Deny Select On dbo.Products to Public;
Grant Select On dbo.vProducts to Public;
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Deny Select On dbo.Employees to Public;
Grant Select On dbo.vEmployees to Public;

Deny Select On dbo.Inventories to Public;
Grant Select On dbo.vInventories to Public;
Go

-- Question 3 (10% pts): How can you create a view to show a list of Category and Product names,
-- and the price of each product?
-- Order the result by the Category and Product!

Create View dbo.vProductsByCategories AS
       Select Top 100000
            C.CategoryName,
            P.ProductName,
            P.UnitPrice
       From Products AS P Join Categories AS C
            On P.CategoryID = C.CategoryID
       Order By C.CategoryName, P.ProductName;
Go

-- Question 4 (10% pts): How can you create a view to show a list of Product names
-- and Inventory Counts on each Inventory Date?
-- Order the results by the Product, Date, and Count!

Create View dbo.vInventoriesByProductsByDates AS
       Select Top 100000
            P.ProductName,
            I.InventoryDate,
            I.[Count]
       From Inventories AS I Join Products AS P
            On I.ProductID = P.ProductID
       Order By P.ProductName, I.InventoryDate, I.[Count];
Go

-- Question 5 (10% pts): How can you create a view to show a list of Inventory Dates
-- and the Employee that took the count?
-- Order the results by the Date and return only one row per date!

-- Here is are the rows selected from the view:

-- InventoryDate    EmployeeName
-- 2017-01-01       Steven Buchanan
-- 2017-02-01       Robert King
-- 2017-03-01       Anne Dodsworth

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Create View dbo.vInventoriesByEmployeesByDates AS
       Select Distinct Top 100000
            I.InventoryDate,
            [EmployeeName] = E.EmployeeFirstName + ' ' + E.EmployeeLastName
       From Inventories AS I Join Employees AS E
            On I.EmployeeID = E.EmployeeID
       Order By I.InventoryDate;
Go

-- Question 6 (10% pts): How can you create a view show a list of Categories, Products,
-- and the Inventory Date and Count of each product?
-- Order the results by the Category, Product, Date, and Count!

Create View dbo.vInventoriesByProductsByCategories AS
       Select Top 100000
            C.CategoryName,
            P.ProductName,
            I.InventoryDate,
            I.[Count]
       From Products AS P Join Categories AS C
            On P.CategoryID = C.CategoryID
            Join Inventories AS I
            On I.ProductID = P.ProductID
       Order By C.CategoryName, P.ProductName, I.InventoryDate, I.[Count];
Go

-- Question 7 (10% pts): How can you create a view to show a list of Categories, Products,
-- the Inventory Date and Count of each product, and the EMPLOYEE who took the count?
-- Order the results by the Inventory Date, Category, Product and Employee!

Create View dbo.vInventoriesByProductsByEmployees AS
       Select Top 100000
            C.CategoryName,
            P.ProductName,
            I.InventoryDate,
            I.[Count],
            [EmployeeName] = E.EmployeeFirstName + ' ' + E.EmployeeLastName
       From Products AS P Join Categories AS C
            On P.CategoryID = C.CategoryID
            Join Inventories AS I
            On I.ProductID = P.ProductID
            Join Employees AS E
            On E.EmployeeID = I.EmployeeID
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            Order By I.InventoryDate, C.CategoryName, P.ProductName, [EmployeeName];
Go

-- Question 8 (10% pts): How can you create a view to show a list of Categories, Products,
-- the Inventory Date and Count of each product, and the Employee who took the count
-- for the Products 'Chai' and 'Chang'?

Create View dbo.vInventoriesForChaiAndChangByEmployees AS
       Select
            C.CategoryName,
            P.ProductName,
            I.InventoryDate,
            I.[Count],
            [EmployeeName] = E.EmployeeFirstName + ' ' + E.EmployeeLastName
       From Products AS P Join Categories AS C
            On P.CategoryID = C.CategoryID
            Join Inventories AS I
            On I.ProductID = P.ProductID
            Join Employees AS E
            On E.EmployeeID = I.EmployeeID
       Where P.ProductName IN ('Chai', 'Chang');
Go

-- Question 9 (10% pts): How can you create a view to show a list of Employees and the Manager who manages them?
-- Order the results by the Manager's name!

Create View dbo.vEmployeesByManager AS
       Select Top 100000
            [Manager] = M.EmployeeFirstName + ' ' + M.EmployeeLastName,
            [Employee] = E.EmployeeFirstName + ' ' + E.EmployeeLastName
       From Employees AS E Join Employees M
            On E.ManagerID = M.EmployeeID
       Order By [Manager], [Employee];
Go

-- Question 10 (20% pts): How can you create one view to show all the data from all four
-- BASIC Views? Also show the Employee's Manager Name and order the data by
-- Category, Product, InventoryID, and Employee.

Create View dbo.vInventoriesByProductsByCategoriesByEmployees AS
       Select Top 100000
            C.CategoryID,
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            C.CategoryName,
            P.ProductID,
            P.ProductName,
            P.UnitPrice,
            I.InventoryID,
            I.InventoryDate,
            I.[Count],
            E.EmployeeID,
            [Employee] = E.EmployeeFirstName + ' ' + E.EmployeeLastName,
            [Manager] = M.EmployeeFirstName + ' ' + M.EmployeeLastName
        From Products AS P Left Join Categories AS C
            On P.CategoryID = C.CategoryID
            Left Join Inventories AS I
            On P.ProductID = I.ProductID
            Left Join Employees AS E
            On I.EmployeeID = E.EmployeeID
            Left Join Employees AS M
            On E.ManagerID = M.EmployeeID
        Order By C.CategoryID, P.ProductID, I.InventoryID, E.EmployeeID;
Go

-- Test your Views (NOTE: You must change the your view names to match what I have below!)
Print 'Note: You will get an error until the views are created!'
Select * From [dbo].[vCategories]
Select * From [dbo].[vProducts]
Select * From [dbo].[vInventories]
Select * From [dbo].[vEmployees]

Select * From [dbo].[vProductsByCategories]
Select * From [dbo].[vInventoriesByProductsByDates]
Select * From [dbo].[vInventoriesByEmployeesByDates]
Select * From [dbo].[vInventoriesByProductsByCategories]
Select * From [dbo].[vInventoriesByProductsByEmployees]
Select * From [dbo].[vInventoriesForChaiAndChangByEmployees]
Select * From [dbo].[vEmployeesByManager]
Select * From [dbo].[vInventoriesByProductsByCategoriesByEmployees]

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